

Muon Lifetime Report

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1 Introduction

The purpose of this experiment is to measure the lifetime of the muon, an unstable elementary particle produced in the upper atmosphere as a result of cosmic ray interactions. Muons are generated primarily through the decay of charged pions and, despite having a short lifetime on the order of microseconds, can be detected at Earth's surface because of relativistic time dilation. Measuring the muon lifetime therefore provides an experimental test of exponential decay and offers insight into weak interactions governing particle decay.

In this experiment, muons incident on a detector assembly are slowed and stopped within a slab of aluminum. The subsequent decay of these muons produces electrons or positrons, which are detected using scintillation detectors coupled to photomultiplier tubes. By measuring the time interval between the arrival of a muon and the detection of its decay product, a distribution of decay times is obtained. Statistical analysis of this distribution allows for the extraction of the muon lifetime.

2 Theory

Muon Decay and Exponential Law

The muon is a lepton with mass approximately 207 times that of the electron and decays via the weak interaction. A positive muon decays according to

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu, \quad (1)$$

while a negative muon decays via

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu. \quad (2)$$

The decay of unstable particles is governed by a first-order differential equation,

$$\frac{dN}{dt} = -\frac{N}{\tau}, \quad (3)$$

where $N(t)$ is the number of undecayed particles at time t , and τ is the mean lifetime. Solving this equation yields the exponential decay law

$$N(t) = N_0 e^{-t/\tau}, \quad (4)$$

where N_0 is the initial number of particles. The probability distribution of decay times follows the same functional form, so a histogram of measured decay times should exhibit an exponential decrease.

Double Exponential Decay Model

In practice, the observed decay distribution deviates from a single exponential because the detected cosmic ray muon population consists of both positive and negative muons. While positive muons decay purely through the weak interaction, negative muons may also undergo nuclear capture when stopped in matter. This additional process increases the effective disappearance rate for negative muons, resulting in a shorter apparent lifetime.

As a result, the observed time distribution is better described by a sum of two exponential components:

$$N(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2}, \quad (5)$$

where A_1 and A_2 are amplitudes, and τ_1 and τ_2 are the characteristic lifetimes of the two decay components.

Additionally, random coincidences between unrelated detector signals introduce a background contribution that is approximately constant in time. Therefore, the full fitting model is

$$N(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + C, \quad (6)$$

where C represents the constant background level.

Statistical Considerations

The number of decay events recorded in each time bin follows Poisson statistics. For a bin with N counts, the associated uncertainty is

$$\sigma = \sqrt{N}. \quad (7)$$

These uncertainties were used to weight the data during the fitting procedure.

3 Equipment and Setup

The experimental setup consists of scintillation detectors, photomultiplier tubes (PMTs), discriminator circuits, and a field-programmable gate array (FPGA) used for timing measurements. A slab of aluminum is placed between detectors to serve as a stopping medium for incident muons.

Three plastic scintillation detectors, labeled A, B, and C, are coupled to photomultiplier tubes, which convert light pulses generated by charged particles into electrical signals. These signals are passed through amplifier/discriminator units, which filter noise and produce standardized digital pulses when the input exceeds a threshold voltage.

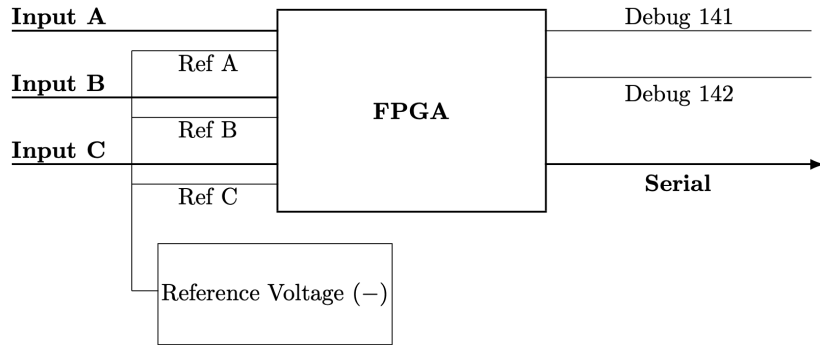


Figure 1: Schematic diagram of the FPGA board.

The outputs of the discriminator channels are connected to an FPGA board, which performs coincidence logic and measures the time interval between detected events. The FPGA records the time difference between a “start” signal, corresponding to a muon entering and stopping in the detector system, and a “stop” signal, corresponding to the detection of the decay electron or positron. These time intervals are output as hexadecimal values through a serial interface to a computer.

Additional equipment includes a high-voltage power supply for the photomultiplier tubes, an oscilloscope for monitoring signals and adjusting thresholds, a function generator for calibration of the timing system, and a computer for data acquisition and analysis.

4 Procedure

Detector Setup and Calibration

The photomultiplier tubes were supplied with high voltage and adjusted to operate near the plateau region of their response curves in order to maximize the signal-to-noise ratio. The discriminator thresholds were set to ensure reliable detection of true particle signals while minimizing noise contributions. Signals from each detector were monitored using an oscilloscope to verify proper operation and timing characteristics.

The FPGA was configured so that the “start” signal was defined by the coincidence between detectors A and B, indicating the passage of a muon that stops within the aluminum slab. The “stop” signal was defined by a pulse from detector C, corresponding to the detection of the decay electron or positron. The FPGA was then configured to measure the time interval between the start and stop signals, recording the result in discrete timing units called ticks.

Data Collection

Following calibration, the detector outputs were reconnected to the FPGA, and data collection was initiated. Several data sets were collected, including three standard runs each spanning one week and one longer run spanning two weeks. This extended acquisition time was necessary to accumulate a statistically significant number of muon decay events. The FPGA output, consisting of hexadecimal values, was recorded and stored for analysis.

5 Data and Calculations

Raw Data and Conversion

The data collected from the FPGA consisted of hexadecimal values representing the time interval between the “start” signal and the “stop” signal. Each hexadecimal value corresponds to a number of discrete FPGA timing units, or ticks.

The raw data were first converted from hexadecimal to decimal using Python:

$$t_{\text{ticks}} = \text{int}(\text{hex value}, 16). \quad (8)$$

The nominal FPGA timing interval was approximately 80 ns per tick, so the decay time was initially computed as

$$t = t_{\text{ticks}} \times 80 \text{ ns}. \quad (9)$$

This conversion factor was verified experimentally using known pulse separations generated by a function generator. Small deviations from the nominal value were incorporated into the systematic uncertainty of the timing calibration.

Histogram Construction

To approximate the decay time distribution $N(t)$, a histogram of the decay times was constructed. The number of bins was chosen using

$$N_{\text{bins}} \approx \sqrt{N_{\text{total}}}, \quad (10)$$

where N_{total} is the total number of recorded events. Each bin represents the number of decay events occurring within a specified time interval, providing a discrete approximation to the continuous decay distribution.

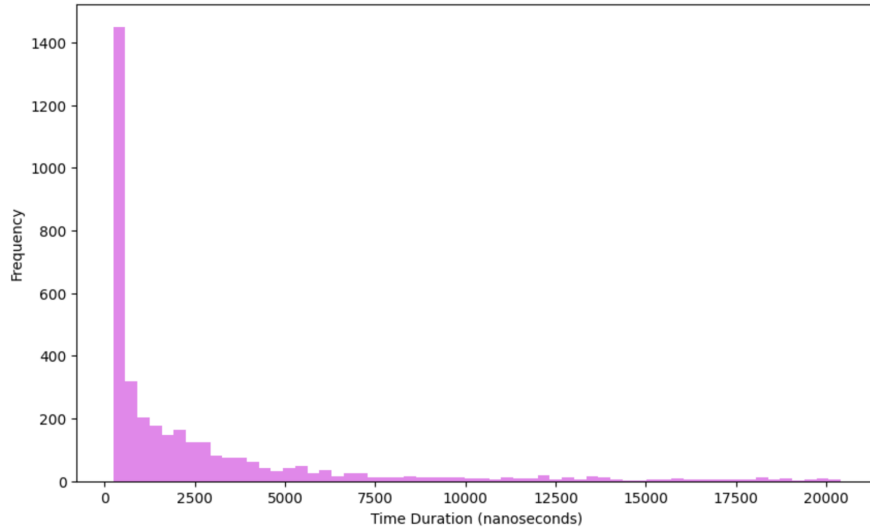


Figure 2: Muon lifetime histogram.

The histogram exhibits the expected exponential-like decay behavior, with a large number of short decay times and progressively fewer long decay events.

Exponential Decay Model

The decay of muons is expected to follow an exponential distribution given by

$$N(t) = N_0 e^{-t/\tau}, \quad (11)$$

where τ is the mean lifetime. However, due to the presence of both positive and negative muons, as well as background noise, a more complete model was used:

$$N(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + C. \quad (12)$$

Both single and double exponential models were fit to the histogram in order to determine which model best described the data. Because the decay model is exponential, statistical fluctuations in the extreme regions of the histogram can significantly affect the quality of the fit. Very small time values may contain electronic noise or timing artifacts, while very large time values contain relatively few events and therefore have larger statistical uncertainty. Excluding a small number of points from the extremities of the dataset is therefore a reasonable approach for improving fit stability and reducing the influence of statistical outliers. Different fitting boundaries were explored to determine the region that produced the most physically consistent and statistically stable fit.

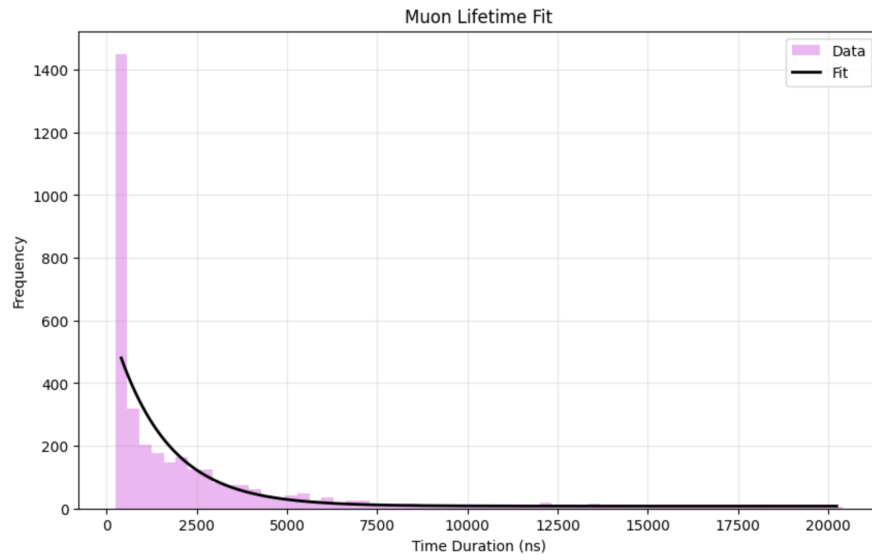


Figure 3: Muon lifetime histogram with single exponential fit.

While the single exponential model captures the general decay trend, noticeable deviations remain throughout the distribution.

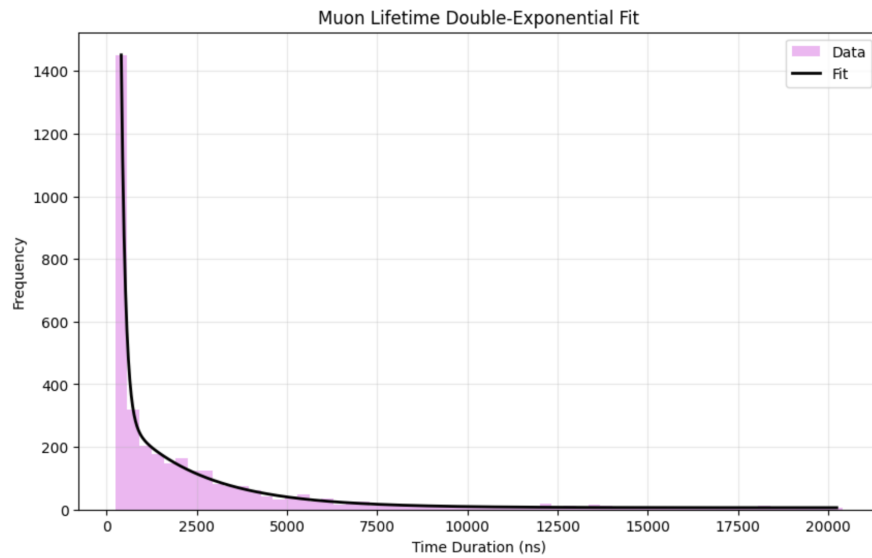


Figure 4: Muon lifetime histogram with double exponential fit.

The dual exponential model provides substantially improved agreement

with the measured distribution, particularly at short decay times. Comparison of the two fitting models demonstrated that the dual exponential model provided better agreement with the measured decay distribution.

Double Exponential Fitting Process

A single exponential model did not adequately describe the observed decay distribution because the detected cosmic ray muon population consisted of both positive and negative muons. Positive muons decay solely through the weak interaction, while negative muons may additionally undergo nuclear capture within the aluminum stopping medium. This introduces two distinct effective decay processes with different characteristic decay times.

To account for these effects, the decay histogram was fit using a double exponential model with a constant background term:

$$N(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + C. \quad (13)$$

The fitting procedure was performed using a weighted non-linear least squares fit. The histogram bin counts were weighted according to Poisson statistics, with uncertainties given by

$$\sigma_i = \sqrt{N_i}. \quad (14)$$

Initial parameter guesses were supplied to the fitting algorithm in order to ensure convergence toward physically meaningful solutions. The fitting algorithm iteratively adjusted the amplitudes, decay constants, and background level to minimize the weighted residuals between the measured histogram and the model.

The resulting fit produced two characteristic decay times:

- τ_1 , corresponding primarily to the longer positive-muon decay component,
- τ_2 , corresponding to the shorter effective negative-muon disappearance time due to both decay and nuclear capture.

Since the accepted free-space muon lifetime is approximately $2.2 \mu s$, the larger of the two fitted decay constants was identified as the physical positive muon lifetime. The shorter component represents the effective disappearance

time of negative muons in aluminum and was therefore not used as the final reported value of the muon lifetime.

Thus, the experimentally determined muon lifetime was taken to be

$$\tau_\mu = \tau_1, \quad (15)$$

where τ_1 was the longer decay constant obtained from the double exponential fit. The uncertainty in the reported lifetime was determined from the covariance matrix returned by the fitting routine:

$$\sigma_\tau = \sqrt{\text{Cov}(\tau_1, \tau_1)}. \quad (16)$$

Comparison of Single and Double Exponential Fits

For the single exponential fit, the experimental muon lifetime was found to be

$$\tau_{\text{single}} = (1.481 \pm 0.04) \mu s. \quad (17)$$

This value is significantly below the accepted muon lifetime of approximately $2.2 \mu s$, indicating that the single exponential model does not fully capture the physics of the observed decay distribution.

For the double exponential fit, the positive muon decay term gave

$$\tau_\mu = (2.179 \pm 0.09) \mu s. \quad (18)$$

This value is consistent with the accepted muon lifetime of approximately $2.2 \mu s$. Therefore, the comparison between the single and double exponential fits justifies the use of the dual exponential model.

6 Discussion and Error Analysis

The experimentally measured muon lifetime was found to be reasonably consistent with the accepted value of approximately $2.2 \mu s$. The double exponential model produced a substantially better fit to the data than the single exponential model, supporting the interpretation that both positive and negative muons contributed to the observed decay distribution.

Several sources of uncertainty contributed to the experimental error. Statistical uncertainty arose from the finite number of recorded decay events and

was treated using Poisson statistics. Systematic uncertainty originated primarily from the timing calibration of the FPGA system, uncertainties in discriminator thresholds, pulse width variations, and accidental coincidence events.

Random accidental coincidences occur independently of the muon decay process and therefore contribute an approximately constant background in the decay-time histogram. This justified the inclusion of the constant background term in the fitting model.

Additional deviations may have resulted from imperfect detector efficiency, electronic noise within the photomultiplier tubes, and fluctuations in the cosmic ray flux during the long acquisition period.

The systematic uncertainty in the timing calibration was estimated by comparing FPGA timing outputs to known pulse separations generated by the function generator. Deviations between the measured and expected timing intervals were used to estimate the uncertainty in the conversion factor between FPGA ticks and physical time.

The accuracy of the experiment could be improved by increasing the total data acquisition time, thereby reducing statistical uncertainty. More precise timing calibration and improved shielding from electronic noise would further reduce systematic error. Improvements in detector efficiency and coincidence timing resolution would also improve the quality of the measured decay distribution.

Because the stop signal was generated using detector C, its discriminator pulse width was chosen to be longer than those of detectors A and B. This ensured that timing jitter and slight electronic delays would not cause valid decay events to be missed by the FPGA coincidence logic.